

An Exact Conjugation Identity for the Many-Body Wilson Loop Beyond Quantization

Kai Watanabe^{1,*}

¹*Independent Researcher*

(Dated: April 14, 2026)

Constraints on the unquantized many-body holonomy are less explored than their quantized counterparts. Here we realize an unquantized regime by tuning the bond dimerization δ and the staggered potential Δ in a dimerized staggered Hubbard ring at half filling. For the tuned parameter sets, a finite excitation gap persists along the $U(1)$ twist cycle $\theta \in [0, 2\pi]$, so that the ground state $|\psi_\delta(\theta)\rangle$ is separated from the excited states. The many-body Wilson loop is therefore well defined from the ground-state family $\{|\psi_\delta(\theta)\rangle; \theta \in [0, 2\pi]\}$. In this setup, we show an exact many-body Wilson loop conjugation identity, $W(-\delta) = W(\delta)^*$, accumulated along a cycle parametrized by θ . Importantly, the identity persists in regimes where the Berry phase $\gamma \equiv -\arg W$ varies continuously. We demonstrate the identity numerically using the density-matrix renormalization group (DMRG) method. The identity extends to other models where the flux-threaded ground-state family along the closed θ -cycle is mapped to the reversed cycle. More generally, the identity can be viewed as a Wilson-loop-level constraint that contains the Berry phase pinning as a fixed-point corollary. Beyond its conceptual content, the identity provides a symmetry-based consistency check for numerical evaluations of Berry phases in interacting systems. It also justifies the signal-to-noise ratio improvement in Monte Carlo simulations by performing simulations at both δ and $-\delta$ and averaging $W(\delta)$ with $W(-\delta)^*$.

I. INTRODUCTION

The many-body Wilson loop, defined from overlaps of an eigenstate family along a $U(1)$ flux-threading cycle, provides a compact characterization of topological many-body systems. It is directly related to physical observables such as Berry phase, polarization and pumped charge [1–3]. Most existing discussions focus on symmetry-enforced quantized regimes, where the observables can be interpreted in terms of topological invariants such as Chern numbers and winding numbers [4].

However, quantization is not the only useful symmetry constraint on many-body Wilson loops. For example, Shiozaki gave a group-theoretical treatment that describes how Wilson loops transform under continuous contour deformations without assuming quantization [5].

In this work, we show an exact Wilson loop identity beyond quantization in an interacting lattice model. In particular, in a dimerized lattice model with a dimerization δ , the exact conjugation identity reads

$$W(-\delta) = W(\delta)^*. \quad (1)$$

As a corollary, the Berry phase $\gamma(\delta) \equiv -\arg W(\delta)$ satisfies $\gamma(-\delta) = -\gamma(\delta) \pmod{2\pi}$.

Crucially, Eq. (1) holds without quantization of $\gamma(\delta)$. (We demonstrate this explicitly in gapped but unquantized regimes realized by tuning a Rice–Mele-type staggered potential together with the bond dimerization in later sections.) Accordingly, Eq. (1) should be viewed as a Wilson-loop-level constraint beyond quantization. Equivalently, Eq. (1) fixes the even/odd parity of $\text{Re } W(\delta)$ and $\text{Im } W(\delta)$ under $\delta \rightarrow -\delta$, providing a direct and gauge-invariant signature of the composite symmetry.

Our result has two immediate implications: (i) it provides a compact organizing principle for the $\delta \rightarrow -\delta$ dependence of the interacting system, and (ii) it serves as a stringent diagnostic for twist-based holonomy computations.

While we mainly concern the dimerized model, the same microscopic logic extends to models in which the permutations of the hopping parameters are equivalent to reversing the bond pattern.

II. CONCRETE EXAMPLE FROM A DIMERIZED STAGGERED HUBBARD MODEL

We consider a $U(1)$ flux θ -threaded dimerized staggered Hubbard ring with even system size L at half filling [6, 7]. The Hamiltonian reads $\hat{H}(\delta, \theta) = \hat{H}_t(\delta, \theta) + \hat{H}_\Delta + \hat{H}_{\text{corr}}$, where

$$\hat{H}_t(\delta, \theta) = \sum_{j=1}^L \sum_{\sigma} \left[\{1 + (-1)^j \delta\} e^{i\theta/L} c_{j\sigma}^\dagger c_{(j+1)\sigma} + \text{H.c.} \right], \quad (2)$$

$$\hat{H}_\Delta = \Delta \sum_{i=1}^L (-1)^i n_i, \quad \left(n_i = \sum_{\sigma} n_{i\sigma} \right), \quad (3)$$

and

$$\hat{H}_{\text{corr}} = U \sum_{i=1}^L n_{i\uparrow} n_{i\downarrow}. \quad (4)$$

Here $c_{i\sigma}$ annihilates a fermion with spin σ on site i . The number operator reads $n_{i\sigma} = c_{i\sigma}^\dagger c_{i\sigma}$. Periodic boundary condition (PBC) $c_{(L+1)\sigma} \equiv c_{1\sigma}$ is imposed in this work. For numerical convenience, we employ a parametrization $\Delta = U/2 - \lambda$. The twist $\theta \in [0, 2\pi]$ is introduced in a

* kaiw.sshrm@gmail.com

uniform gauge, i.e., the same $e^{i\theta/L}$ is attached to every nearest-neighbor hopping. Throughout we work in the fixed-number sector $N_\uparrow = N_\downarrow = L/2$. In this setting, reversal of the bond dimerization $\delta \rightarrow -\delta$ implements the bond-pattern reversal by interchanging even and odd links. We denote by $|\Psi_\delta(\theta)\rangle$ the many-body ground state of $\hat{H}(\delta, \theta)$ in this sector.

This model admits regimes with continuously varying $\gamma(\delta)$ at finite δ while the excitation gap remains finite. This is because, by varying δ and Δ simultaneously, one can change the phase without crossing $\delta = \Delta = 0$ and without closing the gap [8].

Composite particle-hole-translation mapping and the Wilson-loop identity

We show that there exists a composite antiunitary mapping Ξ which maps the flux-threaded ground-state family along the θ cycle to the reversed family:

$$|\Psi_{-\delta}(-\theta)\rangle = e^{i\phi(\theta)} \Xi |\Psi_\delta(\theta)\rangle, \quad (5)$$

where $\phi(\theta)$ is a physically irrelevant θ -dependent phase that cancels on the closed θ -cycle, as we show below. Since $|\Psi_{\pm\delta}(\pm\theta)\rangle$ are eigenstates of $\hat{H}(\pm\delta, \pm\theta)$, it suffices to establish

$$\Xi \hat{H}(\delta, \theta) \Xi^{-1} = \hat{H}(-\delta, -\theta). \quad (6)$$

For this model, Ξ can be constructed as the product of an one-site translation T_1 , a particle-hole transformation $\mathcal{C}_{\text{ph}}$, and a complex conjugation $\mathcal{K}$:

$$\Xi \equiv \mathcal{K} \mathcal{C}_{\text{ph}} T_1. \quad (7)$$

The one-site translation acts as $T_1 c_{i\sigma} T_1^{-1} = c_{(i+1)\sigma}$ and $T_1 n_{i\sigma} T_1^{-1} = n_{(i+1)\sigma}$, where the PBC ensures the well definedness. The particle-hole transformation reads $\mathcal{C}_{\text{ph}} c_{i\sigma} \mathcal{C}_{\text{ph}}^{-1} = (-1)^i c_{i\sigma}^\dagger$. Simple algebra yields $\mathcal{C}_{\text{ph}} n_{i\sigma} \mathcal{C}_{\text{ph}}^{-1} = 1 - n_{i\sigma}$.

The composite mapping Ξ acts on Hamiltonian $\hat{H}(\delta, \theta)$ as follows. First, under T_1 the signs on the staggered factor and the dimerized bond flip as $(-1)^i \mapsto -(-1)^i$. Thus, $\{1 + (-1)^i \delta\} \mapsto \{1 - (-1)^i \delta\}$ equivalent to $\delta \mapsto -\delta$. Similarly, $\hat{H}_\Delta$ changes sign under T_1 . Second, $\mathcal{C}_{\text{ph}}$ flips the sign of $\hat{H}_\Delta$ again, because $n_i \mapsto 2 - n_i$ and $\sum_i^{L=\text{even}} (-1)^i = 0$, so that $\mathcal{C}_{\text{ph}} \hat{H}_\Delta \mathcal{C}_{\text{ph}}^{-1} = -\hat{H}_\Delta$. The on-site correlation term $\hat{H}_{\text{corr}} = U \sum_i n_{i\uparrow} n_{i\downarrow}$ is invariant under $\mathcal{C}_{\text{ph}}$ up to an additive constant proportional to $L - N_{\text{tot}}$, which vanishes at half filling. Finally, complex conjugation $\mathcal{K}$ flips the U(1) flux as $e^{+i\theta/L} \mapsto e^{-i\theta/L}$, i.e., $\theta \mapsto -\theta$. Combining these actions, we obtain the mapping Eq. (6).

The many-body Wilson-loop is defined as

$$W(\delta) = \prod_{j=0}^{N_\theta-1} \mathcal{O}_j(\delta; \theta), \quad (8)$$

where $\mathcal{O}_j(\delta; \theta) = \langle \Psi_\delta(\theta_j) | \Psi_\delta(\theta_{j+1}) \rangle$ is the overlap. The θ -grid is defined as $\theta_j = \frac{2\pi j}{N_\theta}$ and $\theta_{N_\theta} \equiv \theta_0$. Eq. (5) implies overlaps transform by complex conjugation, $\mathcal{O}_j(-\delta; -\theta) = e^{i\phi(\theta_{j+1}) - i\phi(\theta_j)} \mathcal{O}_j^*(\delta; \theta)$. Taking the product over j along the closed twist cycle $\theta_{N_\theta} \equiv \theta_0$, the additional phases cancel. Hence, the Wilson-loop obeys the exact conjugation identity

$$W(-\delta) = W(\delta)^*.$$

III. NUMERICAL DEMONSTRATION

Representative setup and numerical stability

In this work, the ground states $|\Psi_\delta(\theta)\rangle$ are computed by the density-matrix renormalization group (DMRG) [9, 10].

In practice, we set the parameters (U, λ, δ) to a region where the charge and excitation gaps remain finite, though small. In this regime, the two-parameter deformation space spanned by (δ, λ) allows gapped continuous deformations, so that $\gamma(\delta)$ is not pinned and can vary continuously. Unless otherwise stated, we fix $\lambda = 0.67$ and focus on two representative interaction strengths, $U = 6.0$ and $U = 7.0$. We set the dimerization to $\delta = \pm 0.005$ and ± 0.01 . With these parameters, the Berry phase is continuous (depinned). We also include $\delta = 0$ as a pinned reference case.

Throughout the parameter sets used in this study, the excitation gap $\Delta E(\theta_j) = E_1(\theta_j) - E_0(\theta_j)$ attains its minimum at $j = 0$ and remains finite there. From this observation, we conclude that the gap is bounded from below, $\min_j \Delta E(\theta_j) > 7.0 \times 10^{-2}$. Moreover, the low-lying energies $E_n(\theta_j)$ show negligible dependence on the discretization N_θ over the resolutions used in our calculations.

We assess the reliability of the overlap Wilson loop W by explicitly tracing the link overlaps $\{\mathcal{O}_j(\delta; \theta)\}$ along the twist cycle. Throughout the parameter sets studied, the overlaps are well controlled, with $|\mathcal{O}_j(\delta; \theta)| \gtrsim 0.97$ except for the closing link $\theta_{N_\theta-1} \rightarrow \theta_0$ given by

$$\mathcal{O}_{\text{cl}}(\delta; \theta) \equiv \langle \Psi_\delta(\theta_{N_\theta-1}) | \Psi_\delta(\theta_0) \rangle.$$

This link in the θ -grid gives the minimum overlap, $|\mathcal{O}_{\text{cl}}| = \min_j |\mathcal{O}_j|$, and typically lies in the range 0.1–0.5 for the sets of δ , λ and U used in this study. Hence the closing overlap dominantly controls the suppression of $|W(\delta)|$.

As also reported in Ref. [11], we interpret this behavior as a discretization artifact of the finite- N_θ twist cycle, since only the closing link stitches the endpoints $\theta_{N_\theta-1} \approx 2\pi$ and $\theta_0 = 0$. The absolute value of the closing-link overlap decreases with N_θ and converges to a finite value (~ 0.36 for $(\delta, \lambda, U) = (0.01, 0.67, 7.0)$) as N_θ is increased. The artifact becomes less pronounced as $|\delta|$ increases, consistent with Fig. 1 where $|W|$ is smallest at $\delta = 0$ and increases for $\delta = \pm 0.01$.

We also verified that the conjugation identity and the qualitative depinning behavior are insensitive, within the

tested range, to standard DMRG convergence controls (bond dimension and sweep number) as well as to system size, while the extracted value of $\gamma(\delta)$ can depend on the system size L .

Numerical demonstration of the Identity

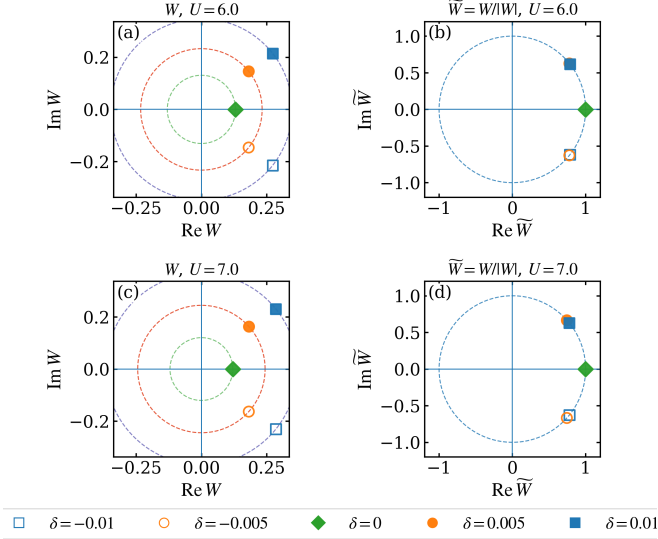


FIG. 1. Complex-plane plots of the many-body Wilson loop $W(\delta)$ and its normalized version $\tilde{W}(\delta) \equiv W(\delta)/|W(\delta)|$ for representative θ -grid $N_\theta = 48$. Panels (a,b) [top row]: $U = 6.0$; panels (c,d) [bottom row]: $U = 7.0$. Markers indicate $\delta \in \{-0.01, -0.005, 0, 0.005, 0.01\}$. Marker shape and color encode $|\delta|$ (shared by $\pm\delta$), and filled (open) markers indicate $\delta > 0$ ($\delta < 0$). Dashed circles are shown as guides to the eye.

Fig. 1 shows the Wilson loop level complex-conjugation relation. For representative dimerizations, $\delta = \pm 0.005$ and $\delta = \pm 0.01$, we find that the many-body Wilson loop satisfies

$$W(-\delta) \simeq W(\delta)^*,$$

within numerical accuracy. (At $\delta = 0$ the Wilson loop is pinned to real value and satisfies the identity as well.)

Since the Berry phase is defined by $\gamma(\delta) \equiv -\arg W(\delta)$, the identity implies $\gamma(-\delta) \simeq -\gamma(\delta)$. Fig. 1 verifies this oddness directly at finite δ in a depinned regime where $\gamma(\delta)$ varies smoothly. This shows that the Wilson loop conjugation identity holds without invoking Berry phase pinning, complementing the symmetry-enforced pinned setting discussed in Ref. [12].

Fig. 2 summarizes the θ -grid N_θ dependence of the Berry phase. We plot $\gamma(\delta)$ evaluated at each N_θ used in this study, shown as a function of $1/N_\theta$. To provide a compact parametrization over the tested range, we fit the data as a function of $1/N_\theta$ by

$$\gamma(\delta; N_\theta) \simeq \gamma_\infty(\delta) + a(\delta)/N_\theta + b(\delta)/N_\theta^2,$$

where we write the Berry phase with an explicit N_θ argument to make the twist-grid dependence explicit. As shown in Fig. 2, the data are well captured by the fitted curve. As a representative example, for $U = 7.0$ and $\delta = 0.01$ we obtain the provisional coefficients

$$\gamma_\infty(0.01) \simeq -0.800, \quad a(0.01) \simeq 6.466, \quad b(0.01) \simeq -22.15,$$

from an unweighted quadratic fit over the tested N_θ values. (For this fit, the unweighted RMSE is $\simeq 3.3 \times 10^{-3}$ over the tested range.)

Consistent with the conjugation constraint and its Berry phase corollary, the fits for $\gamma(\pm\delta; N_\theta)$ form mirrored curves with opposite curvature. The quadratic coefficients approximately flip sign under $\delta \rightarrow -\delta$; for instance, for $\delta = -0.01$ we find $(\gamma_\infty(-0.01), a(-0.01), b(-0.01)) \simeq (0.800, -6.464, 22.14)$.

As an internal large- N_θ check, for $\delta = 0.01$, we also evaluated $\gamma(0.01; N_\theta)$ up to $N_\theta = 512$, finding a residual deviation from the quadratic fit of order $|\gamma(0.01; 512) - \gamma_\infty(0.01)| \sim 10^{-2}$. A fully quantitative characterization of $\gamma_\infty(\delta)$ would require a more systematic finite- N_θ (and finite- L) fitting, or alternatively the use of polarization estimators based on the many-body position operator [3].

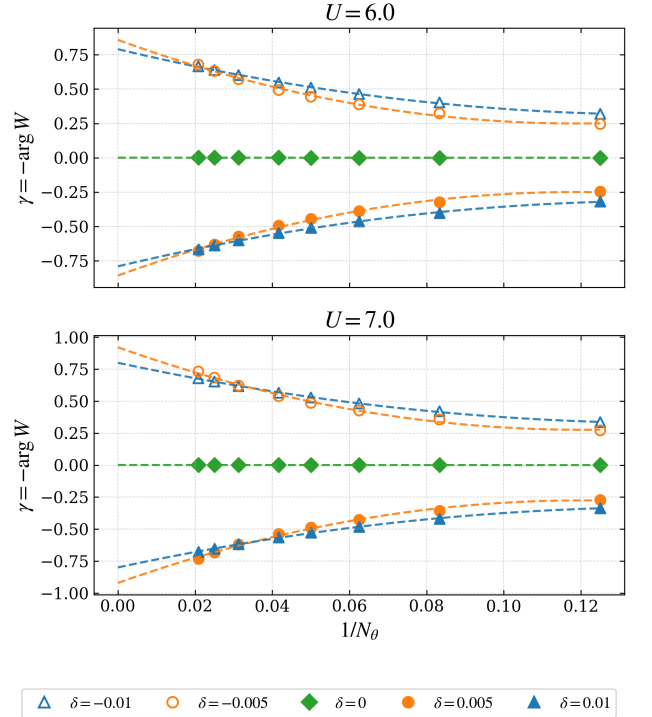


FIG. 2. Berry phase $\gamma(\delta) \equiv -\arg W(\delta)$ plotted versus $1/N_\theta$. Top (bottom) panel: $U = 6.0$ ($U = 7.0$). The data points are for $\delta \in \{-0.01, -0.005, 0, 0.005, 0.01\}$ and $N_\theta \in \{8, 12, 16, 20, 24, 32, 40, 48\}$. Dashed curves indicate quadratic fits in $1/N_\theta$ to each data set. Marker shape and color encode $|\delta|$ (shared by $\pm\delta$), and filled (open) markers indicate $\delta > 0$ ($\delta < 0$).

IV. DISCUSSIONS

The key message of this letter is that the bond-pattern reversal combined with the flux orientation reversal enforces the Wilson loop conjugation identity Eq. (1), which is directly usable in lattice-model calculations as a constraint relating $\pm\delta$. This statement does not rely on any quantization of the Berry phase and remains meaningful in the depinned regime where the Wilson loop is generically complex.

Our formulation provides a complementary bottom-up route: starting from a microscopic mapping of the flux-threaded Hamiltonian (and hence the associated ground-state family), we derive an explicit loop action $(\delta, \theta) \mapsto (-\delta, -\theta)$ and obtain $W(-\delta) = W(\delta)^*$ as a directly usable lattice-model identity.

In this work, the bond-pattern inversion is realized simply by flipping the sign of the dimerization ($\delta \rightarrow -\delta$). This is particular to the dimerized case. However, the same strategy applies to models where the flux-threaded ground-state family along the closed θ -cycle, $\{|\Psi_{\mathbf{t}}(\theta)\rangle\}$, is mapped to the reversed cycle $\{|\Psi_{\mathbf{P}\mathbf{t}}(-\theta)\rangle\}$ up to a physically irrelevant θ -dependent phase, where $\mathbf{P}$ denotes an appropriate permutation of microscopic parameters (e.g., a bond-pattern reversal of the hopping set $\mathbf{t} = \{t_{ij}\}$). This is a generalization of Eq.(5), and, in such cases, the Wilson loop satisfy a similar identity. This also applies to continuum models when an analogous loop-reversal mapping exists; see Supplemental Material [13] for a Kronig–Penney model example. Whenever a nondegenerate ground-state family along the closed θ -cycle is mapped to the reversed cycle by an antiunitary transformation (up to a phase), the corresponding Wilson loop satisfies a conjugation constraint.

Fundamentally, the identity is rooted in the microscopic symmetry of the ground state family $\{|\psi_{\delta}(\theta)\rangle; \theta \in$

$[0, 2\pi]\}$ along the θ -cycle. Naturally, a group-theoretical approach provides an alternative perspective. A related symmetry-based framework is the recent work by Shiozaki, who formulated a group-theoretical description of the response of Wilson loops to continuous deformations of the θ -cycle [5].

Beyond its conceptual content, Eq. (1) provides a practical consistency check for numerical evaluations of interacting holonomies. Deviations from the predicted even/odd structure of $\text{Re } W$ and $\text{Im } W$ under $\delta \rightarrow -\delta$ indicate either explicit breaking of the composite symmetry (e.g., departures from half filling or additional perturbations) or numerical inaccuracies in the overlap-based construction of the Wilson loop. A stringent internal check is provided by

$$\Delta_W(\delta) \equiv |W(-\delta) - W(\delta)^*|.$$

One may use the identity to improve the signal in Monte Carlo simulations. For instance, one can sample both $W(\delta)$ and $W(-\delta)$ and average $W(\delta)$ with $W(-\delta)^*$. In an ideal case where the two samples are uncorrelated, this averaging reduces the noise by a factor of $\sqrt{2}$.

At $\delta = 0$, the identity reduces to

$$\gamma(0) = -\gamma(0) \pmod{2\pi},$$

and hence the Berry phase is pinned to $\gamma(0) \in \{0, \pi\}$. This reproduces the familiar $\mathbb{Z}_2$ quantization of Berry phases (see, e.g., Ref. [12]). Thus, the pinning appears as a fixed-point corollary of the more general Wilson loop conjugation identity, which remains valid away from symmetry-enforced quantization.

In unquantized regimes, such exact identities can provide useful constraints on the many-body holonomy. Exploring further constraints of this type is an interesting direction for future work.

-
- [1] J. Zak, Berry's phase for energy bands in solids, *Phys. Rev. Lett.* **62**, 2747 (1989).
 - [2] R. D. King-Smith and D. Vanderbilt, Theory of polarization of crystalline solids, *Phys. Rev. B* **47**, 1651 (1993).
 - [3] R. Resta, Quantum-mechanical position operator in extended systems, *Phys. Rev. Lett.* **80**, 1800 (1998).
 - [4] C.-K. Chiu, J. C. Y. Teo, A. P. Schnyder, and S. Ryu, Classification of topological quantum matter with symmetries, *Rev. Mod. Phys.* **88**, 035005 (2016).
 - [5] K. Shiozaki, Equivariant parameter families of spin chains: A discrete MPS formulation, *SciPost Physics* **20**, 024 (2026).
 - [6] M. J. Rice and E. J. Mele, Elementary excitations of a linearly conjugated diatomic polymer, *Phys. Rev. Lett.* **49**, 1455 (1982).
 - [7] W. P. Su, J. R. Schrieffer, and A. J. Heeger, Solitons in polyacetylene, *Phys. Rev. Lett.* **42**, 1698 (1979).
 - [8] T. Morimoto and A. Furusaki, Topological classification with additional symmetries from clifford algebras, *Phys. Rev. B* **88**, 125129 (2013).
 - [9] S. R. White, Density matrix formulation for quantum renormalization groups, *Phys. Rev. Lett.* **69**, 2863 (1992).
 - [10] J. Hauschild and F. Pollmann, Efficient numerical simulations with Tensor Networks: Tensor Network Python (TeNPy), *SciPost Phys. Lect. Notes*, 5 (2018).
 - [11] K. Watanabe, Symmetry-enforced agreement of Kohn–Sham and many-body Berry phases in the SSH–Hubbard chain (2026), arXiv:2602.22578 [cond-mat.str-el].
 - [12] Y. Hatsugai, Quantized berry phases as a local order parameter of a quantum liquid, *Journal of the Physical Society of Japan* **75**, 123601 (2006).
 - [13] See Supplemental Material at [https://arxiv.org/abs/2603.22217] for a Kronig–Penney model demonstration of the Wilson loop conjugation identity.